

Interface Capability Brief

FHIR R4 Expandable Platform (Phase 6) · Stakeholder overview · 2026-08-05

1. Executive summary

Expandable HL7 FHIR R4 PilotFish platform through Phase 6 Bulk \$export.

FHIR 4.0.1 · CapabilityStatement 0.6.0 · draft

2. Who this is for

- Business / clinical stakeholders evaluating whether the integration covers needed workflows
- Solution architects comparing PilotFish capabilities to project requirements
- Demo audiences walking the happy-path in the Web UI

3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

FHIR resources & actions

Phase 6 expandable FHIR R4 façade with async system-level Bulk \$export (NDJSON), Keycloak Bearer on writes/export, HAPI validation, search, and Bundle transaction/batch. Deferred: multi-partition Bulk, group export, _since deletion markers, full SMART launch.

Resource	Supported actions	Search highlights
Patient	Create, Read, Update, Delete (soft), Search	_id, identifier, family, given, name, gender, birthdate
Practitioner	Create, Read, Update, Delete (soft), Search	_id, identifier, family, given, name
PractitionerRole	Create, Read, Update, Delete (soft), Search	_id, q
Organization	Create, Read, Update, Delete (soft), Search	_id, identifier, name, active
Location	Create, Read, Update, Delete (soft), Search	_id, q
Encounter	Create, Read, Update, Delete (soft), Search	_id, patient, date, status, class
Observation	Create, Read, Update, Delete (soft), Search	_id, patient, code, date, status, category
Condition	Create, Read, Update, Delete (soft), Search	_id, patient, code, clinical-status, onset-date
Procedure	Create, Read, Update, Delete (soft), Search	_id, q
AllergyIntolerance	Create, Read, Update, Delete (soft), Search	_id, q
MedicationRequest	Create, Read, Update, Delete (soft), Search	_id, q
Medication	Create, Read, Update, Delete (soft), Search	_id, q
Immunization	Create, Read, Update, Delete (soft), Search	_id, q
DiagnosticReport	Create, Read, Update, Delete (soft), Search	_id, q
DocumentReference	Create, Read, Update, Delete (soft), Search	_id, q
CarePlan	Create, Read, Update, Delete (soft), Search	_id, q
CareTeam	Create, Read, Update, Delete (soft), Search	_id, q

Resource	Supported actions	Search highlights
Goal	Create, Read, Update, Delete (soft), Search	_id, q
ServiceRequest	Create, Read, Update, Delete (soft), Search	_id, q
Coverage	Create, Read, Update, Delete (soft), Search	_id, q
Claim	Create, Read, Update, Delete (soft), Search	_id, q
ExplanationOfBenefit	Create, Read, Update, Delete (soft), Search	_id, q
Appointment	Create, Read, Update, Delete (soft), Search	_id, q
Schedule	Create, Read, Update, Delete (soft), Search	_id, q
Slot	Create, Read, Update, Delete (soft), Search	_id, q
RelatedPerson	Create, Read, Update, Delete (soft), Search	_id, q
Person	Create, Read, Update, Delete (soft), Search	_id, q
EpisodeOfCare	Create, Read, Update, Delete (soft), Search	_id, q
Binary	Create, Read, Update, Delete (soft), Search	_id, q
Bundle	Create, Read, Update, Delete (soft), Search	_id, q
Parameters	Create, Read, Update, Delete (soft), Search	_id, q

31 FHIR resource type(s) declared for this instance.

Server operations & system interactions

- \$export — Demo async system Bulk export to NDJSON under /opt/pilotfish/output/bulk-export/{jobId}.
- System interaction: transaction
- System interaction: batch
- Bundle/\$transaction-or-batch

Supported formats: application/fhir+json, json

4. Honest boundaries (not claimed yet)

- Full search grammar / history
- US Core IG packages
- Group / Patient compartment export
- CapStatement \$validate
- Full SMART EHR launch
- Multi-node Bulk, _since tombstones
- multi-partition Bulk, group export, _since deletion markers, full SMART launch.

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Create and read clinical resources (for example Patient) through the FHIR REST façade
- Search using the indexed parameters declared for each resource type
- Reject or accept writes according to validation rules configured for the demo
- Kick off an async Bulk \$export, poll job status, and download NDJSON output files
- Submit a Bundle as transaction (atomic) or batch (per-entry status)

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- Confirm unauthenticated writes are rejected and Bearer tokens unlock write/export paths

6. How it works (plain language)

Jobs + files: /opt/pilotfish/output/bulk-export/{jobId}/ (+ dbo.FhirExportJobs).

Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

- 1 - FHIR R4 REST Platform: runs steps such as FHIR R4 REST Platform, Save Raw Body, Extract Body ResourceType, Extract Logical Id.
- 2 - FHIR Outbound Client: runs steps such as Outbound Request Poll, Extract Outbound URL, Conditional Router, Save Outbound Response Body.

7. Security & trust posture

- POST/PUT/DELETE require Authorization: Bearer <access_token>. GET metadata/read/search are open in this demo.
- Keycloak OAuth2/OIDC (demo SMART façade)
- OAuth2 / OIDC Bearer tokens (Keycloak in this demo) protect write and export operations
- Inbound clinical/FHIR payloads can be validated before persistence

8. Seeing it work

- SQL / 14338
- EIP / 8110
- Web UI / 8111
- Keycloak / 8112
- docker compose up -d --build
- | Capability Brief (shareable) | documents/FHIR_R4_Platform_Capability_Brief.pdf | <http://127.0.0.1:8111/documents/capability-brief.pdf> |
- | Route diagrams (technical) | documents/FHIR_R4_Platform_V2_Route_Diagrams.pdf | <http://127.0.0.1:8111/documents/route-diagrams.pdf> |
- 'http://127.0.0.1:8110/eip/rest/fhir/\$export?_type=Patient,Observation'

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- FHIR_R4_Platform_V2_Route_Diagrams.pdf — detailed PilotFish route wiring diagrams
- eip-root/interfaces/FHIR R4 Platform/routes/1 - FHIR R4 REST Platform/capability-statement.json — machine-readable FHIR CapabilityStatement
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.